

MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

1. Introduction

MedTrack_DV is a healthcare operations analytics project developed using data preparation and Tableau visualization. The project transforms patient and hospital operational data into meaningful analytical dashboards.

The analysis focuses on hospital performance, patient flow, department performance, and resource utilization.

2. End-to-End Methodology

The project follows a structured end-to-end methodology, beginning with data collection and ending with documentation and delivery.

Data Collection

Healthcare data was collected from the Kaggle Hospital HMIS dataset. The dataset contains information related to patients, admissions, departments, wards, beds, billing, diseases, employees, and staff assignments. Relevant tables were integrated to create a consolidated dataset for analysis.

Data Cleaning

The collected data was cleaned to improve its quality and consistency. This process included checking duplicate records, handling missing values, standardizing column names, correcting data types, checking date fields, removing unnecessary columns, and validating relationships between tables. The final dataset contains approximately 45,000 records and 34 columns, with no duplicate records identified.

Data Transformation

The cleaned data was transformed into an analysis-ready format. Relevant healthcare tables were combined, calculated fields were created, admission and discharge information was derived, patients were grouped into age categories, monthly date fields were prepared, admission types were categorized, and department- and ward-level attributes were organized.

KPI Engineering

Key Performance Indicators were created to measure hospital operations and performance. These include Total Admissions, Total Patients, Total Discharges, Average Length of Stay, Readmission Rate, Occupancy Rate, Bed Utilization Rate, Department Efficiency, Total Beds, Total Staff, Total Wards, Average Beds per Ward, and Total Revenue. These KPIs were calculated using aggregation, distinct counts, date calculations, and calculated fields.

Tableau Data Preparation

The prepared dataset was connected to Tableau for visualization. Field names and data types were verified, calculated fields were created, relationships between tables were checked, and KPI values were validated. Dimensions and measures were organized to support the development of interactive dashboards.

Dashboard Development

Four interactive dashboards were developed in Tableau: Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization. Each dashboard was designed to focus on a specific area of healthcare operations and includes relevant KPIs, charts, filters, and visual comparisons.

Dashboard Integration

The dashboards were integrated using navigation buttons and interactive filters. A consistent visual design, common titles, KPI labels, and dashboard navigation were maintained to provide users with a connected and easy-to-use dashboard experience. In Dashboard Integration Below were used.

- 1. Hospital Operations Analysis** – Analyses hospital activities such as admissions, discharges, bed capacity, staff availability and ward utilisation. This helps management understand the overall workload and operational performance of the hospital.
- 2. Patient Flow Analysis** – Examines the movement of patients through the hospital, from admission to discharge. It uses admission dates, discharge dates, admission types and patient information to identify patient-flow patterns and improve hospital service efficiency.
- 3. Length of Stay Analysis** – Calculates the number of days patients stay in the hospital using admission and discharge dates. Average Length of Stay helps evaluate treatment efficiency, patient turnover and bed usage.
- 4. Readmission Analysis** – Identifies patients who have been admitted more than once. The Readmission Rate helps monitor recurring admissions and supports the identification of possible quality-of-care or follow-up issues.
- 5. Bed Occupancy and Utilisation Analysis** – Measures how hospital beds are occupied and utilised across departments and wards. This helps management monitor bed availability, avoid underutilisation and support effective capacity planning.
- 6. Department Performance Analysis** – Compares departments based on admissions, discharges, efficiency, readmission rates, staffing and financial activity. This helps identify high-performing departments and areas requiring improvement.

7. Resource Utilisation Analysis – Analyses the availability and usage of hospital resources such as beds, wards and staff. It supports better resource allocation and helps management plan resources according to operational requirements.

8. Revenue and Billing Analysis – Uses billing information such as total amount, patient payable amount and insurance-covered amount to understand the revenue generated by the hospital and support financial monitoring.

9. Descriptive Analytics – Summarises historical hospital data using KPIs, charts, trends and category-wise comparisons. It helps explain what happened in the hospital during the selected period.

11. Interactive Dashboard Analysis – Uses filters, cross-filtering, navigation buttons, KPIs and visualisations to allow users to explore hospital data dynamically. This helps hospital management make faster and data-driven decisions.

Overall, these healthcare methodologies help monitor hospital performance, understand patient-flow patterns, improve bed and staff utilisation, compare department efficiency, analyse revenue and support better operational decision-making.